

CM0081 Formal Languages and Automata

§ 1.4 Formal Proofs

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Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Proofs by Contradiction and Proofs of Negations

Proof by contradiction
(or *reductio ad absurdum*)

$$\frac{\begin{array}{c} [\neg\beta] \\ \vdots \\ \perp \end{array}}{\beta}$$

Proof of negation (Bauer 2017)

$$\frac{\begin{array}{c} [\beta] \\ \vdots \\ \perp \end{array}}{\neg\beta}$$

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Justifications

$$\frac{\frac{\frac{[\neg\beta] \quad \vdots \quad \perp}{\neg\beta \rightarrow \perp} \text{ (conditional proof)}}{\neg\neg\beta} \text{ } (\neg\alpha := \alpha \rightarrow \perp)}{\beta} \text{ } (\vdash \neg\neg\alpha \rightarrow \alpha)$$

Proof of negation (Bauer 2017)

$$\frac{\begin{array}{c} [\beta] \\ \vdots \\ \perp \end{array}}{\neg\beta}$$

$$\frac{\frac{[\beta] \quad \vdots \quad \perp}{\beta \rightarrow \perp} \text{ (conditional proof)}}{\neg\beta} \text{ } (\neg\alpha := \alpha \rightarrow \perp)$$

Inductive Proofs: Mathematical Induction

The induction principle

Let $S(n)$ be a property about natural numbers. If

- (i) we prove $S(i)$ (basis step) and
 - (ii) we prove that for all natural number $n \geq i$, $S(n)$ implies $S(n + 1)$ (inductive step),
- then we may conclude $S(n)$ for all $n \geq i$.

Inductive Proofs: Structural Induction

The structural induction principle

Let $S(X)$ be a property about structures X that are defined by some recursive/inductive definition. If

- (i) we prove $S(X)$ for the basis structure(s) of X (basis step) and
- (ii) given a structure X whose recursive/inductive definition says it is formed from $Y_1, \dots, Y_k$, we prove $S(X)$ assuming that the properties $S(Y_1), \dots, S(Y_k)$ hold (inductive step),

then $S(X)$ is true for all X .

Inductive Proofs: Mutual Induction

Example

Given the functions $f, g, h : \mathbb{N} \rightarrow \mathbb{N}$,

$$\begin{array}{lll} f(0) = 0, & g(0) = 1, & h(0) = 0, \\ f(n+1) = g(n), & g(n+1) = f(n), & h(n+1) = 1 - h(n), \end{array}$$

and properties R, S, T ,

$$R(n) : h(n) = 1 - g(n), \quad S(n) : h(n) = f(n), \quad T(n) : S(n) \wedge R(n),$$

Inductive Proofs: Mutual Induction

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1. To prove $(\forall n)R(n)$ (impossible!)
2. To prove $(\forall n)S(n)$ (impossible!)

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1. To prove $(\forall n)R(n)$ (impossible!)
2. To prove $(\forall n)S(n)$ (impossible!)
3. To prove $(\forall n)T(n)$ (by mutual induction)

(continued on next slide)

Inductive Proofs: Mutual Induction

Proof

- Basis step $T(0)$. Easy.

Inductive Proofs: Mutual Induction

Proof

- Basis step $T(0)$. Easy.
- Induction step $T(n) \Rightarrow T(n+1)$:

$$\begin{aligned} S(n) : \quad h(n+1) &= 1 - h(n) && \text{(def. of } h) \\ &= g(n) && \text{(IH } R(n)) \\ &= f(n+1) && \text{(def. of } f) \end{aligned}$$

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References

- Andrej Bauer (2017). Five States of Accepting Constructive Mathematics. Bulletin of the American Mathematical Society 54.3, pp. 481–498. DOI: [10.1090/bull/1556](https://doi.org/10.1090/bull/1556) (cit. on pp. 3, 4).
- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).